Analysis of Election Data

COSC5806002 - Data Analysis with Python

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March 5, 2026

Abstract

The main objective of this project is to pre-process, analyze and visualize the US State Election data in a dashboard for the much better understanding of the US political trends. We focused our analysis on the election data from year 1976 to 2020. The team downloaded the US State election data file from the Harvard Dataverse website and performed data cleaning using Python for the accurate results. We created a dashboard which includes efficient visualizations using the tool Tableau. Finally, integrated these visualizations in a web application using React. This dashboards plays a crucial role in forecasting the political patterns of future US elections.

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1 Introduction

The US state election has been a hot topic for many years. The Analysis of US State Elections Results helps us to understand political trends and voter behavior in USA, and individual states. Over the years, US elections have reflected broader political, social, and demographic shifts in the country.

Our project focuses on analyzing US Senate election results from multiple years (1976 2020) to find patterns and trends in the election. The overall goal of the project is not only to analyze the election results but also find political dominance of parties across different states and how they are deviating during the period. It will also contribute to discussions about future electoral patterns and political forecasting in addition to offering an overview of previous elections through data-driven insights and visual narratives.

The main objective of this project will be to explore and learn various data analytics thinking and techniques through python programming and its various libraries. Simultaneously, learning Power BI as well as Tableu for development of interactive dashboards which will give deep understanding of its features for data analysis and visualization. This makes us masters in the practical usage of various python libraries to explore datasets, summarizing key statistics, detecting the past and future trends as well as to identify various logic for effective visualization methods.

2 Background

Elections to the US Senate have a significant impact on national administration and policy. These elections over the years have reflected greater political, demographic, and social changes throughout the country in addition to determining parliamentary majorities. At the state and federal levels, examining past Senate election data provides important insights into voting patterns, party dynamics, and long-term electoral trends.

The development of data analysis tools and the increasing availability of past election records have made it possible to conduct more complex and interactive political analysis. With its vast library of tools for statistical analysis, data preprocessing, and visualization, Python offers a strong basis for carrying out in-depth analyses of election data. Furthermore, Tableau's dynamic and intuitive visualization features enable the development of interactive dashboards that successfully convey insights to a wide audience.

The results of U.S. Senate elections from 1976 to 2020 are analyzed in this project using these tools. Through the use of strong data processing methods and interactive Tableau dashboards, the study seeks to comprehend past changes in voter behavior, identify important electoral trends, and facilitate discussions about political forecasting for the future. The combination of these tools guarantees that the analysis is both accessible and methodologically correct thereby promoting informed discussion in the field of electoral data analysis.

3 Dataset

Our project utilizes the 1976–2020 US Senate (State Level) [7] dataset, which provides a thorough and comprehensive set of election data at the state level covering more than 40 years. The election year, state, names of the candidates, party affiliations, total number of votes, candidate votes and election results are all included in this dataset. We can do a thorough examination of past voting trends, observe trends in political party dominance, and look at how voter preferences have changed over time in various U.S. states by using this dataset. This serves as the basis for producing insightful visualizations and analyses that facilitate both future-oriented political debate and comprehension of the past.

4 Project Implementation

4.1 Technology Stacks

- **Python:** In this project, we used Python[5] as the main programming language due to its extensive support for data analysis and manipulation. Python ecosystem provides numerous libraries and packages for efficient processing of data.

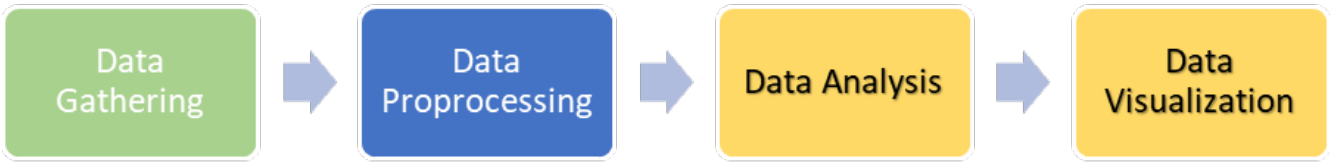

Figure 1: FLowchart of Typical Analysis project

- **Numpy:** To manage numerical data and perform arithmetic operations, we used Numpy. It is a Python library that provides multi-dimensional array and a set of functions for handling numeric data [**numpy**]. Numpy is useful when handling large data-set like election data containing numeric values.
- **Pandas:** Pandas is a fast, flexible and popular tool for data analysis[10]. It provides strong and easy to use data structures such as DataFrame and Series for handling structured datasets. We used pandas to clean, filter, group, sort our data. We also used it to handle missing values.
- **React:** React[8] is a widely used JavaScript / TypeScript based web development framework. We used React in enhancement face of our project to integrate dashboard created using Tablue.
- **Django:** Django[4] is a popular framework for creating web applications such as dashboards in python. We used Django to fetch the election dashboard details from the database.
- **Postgres SQL:** Postgres[6] is a powerful, robust open source relational database system which work well with Django. We used Postgres to store the election dashboard details.

4.2 Tools and Services

- **Tableu:**
  Tablue[11] is a very user friendly tool that helps in data visualizations and dashboard creation. Our team used Tablue to create an interactive election dashboard with meaningful visualizations.
- **Docker**
  Docker is a powerful opersource platform that make building running and shipping applications. Pre-build postgres [3] image from the docker hub is used to run the database in development environment.
- **Neon**
  Neon[9] is a serverless postgres database that built for the cloud environment. This is a perfect tool for developers who use Django. We used this tool to host out data base.
- **Vercel:**
  Vercel is a cloud platform that used for deploying the front end applications that built on the framework like React. This was the platform for us to host our react application for dashboard.
- **Railway**
  Railway[2] is a developer friendly cloud platform to deploy full-stack apps that helps to keep backend, front end and databases in one place. We used it as a platform to host Django.

4.3 Data Preprocessing Analysis

Data Preprocessing is very crucial for data analysis project. It reduces the probability of error or possibility of producing faulty result which could let to misinterpretation of data in the further stage. Data preprocessing involves transforming raw data into a clean, consistent, and usable format for analysis, modeling, or machine learning, by addressing issues like missing values, inconsistencies, and incorrect formats.

Data preprocessing is usually done after the initial stage of Data Gathering in a typical Data Analysis project development, which act input to further stage like Data Analysis and Data Visualization.

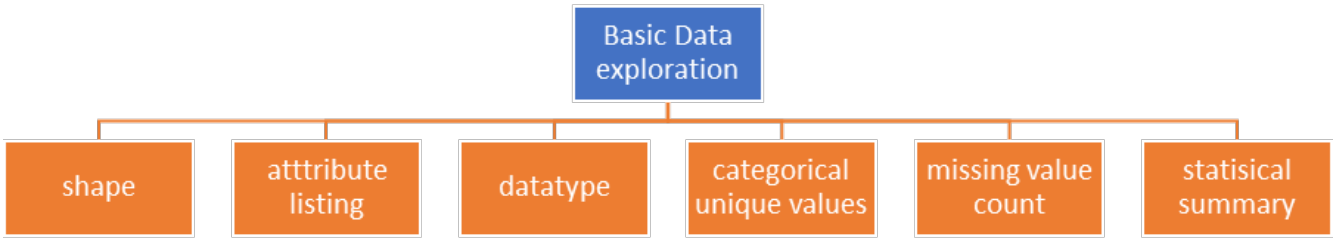

Figure 2: Data Preprocessing Exploratory Data Analysis

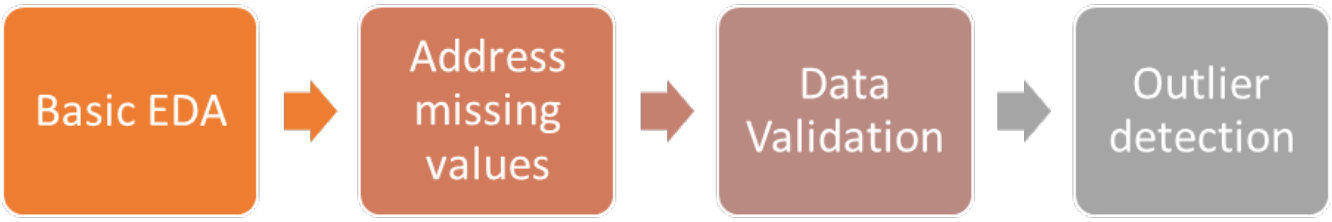

Figure 3: Data Preprocessing Workflow

In this project, we did data preprocessing using **python programming** and we use **Google Collab IDE** which allows collaborative development. For Data preprocessing we use various **python libraries like NumPy and Pandas.**. Under Data preprocessing we did **basic data exploration** which include exploring the shape of dataset, attribute listing, checking the datatype of each feature, number of unique values in each categorical feature, identifying number of missing values, statistical summary of the dataset.

For details preprocessing we did analyse all the attributes in details using the external sources, that is individually as well as its association with other attributes. Based on our studies we address missing value.

In our raw data, we found two attributes with missing values which was candidate and party_detailed, we address those missing by replacing with 'Not Given' values. Because both attributes have existed individually, and they have no association with other attributes and no relation within their own attribute. Lastly 'Not Given' is value which convey a specific meaning.

After addressing the missing values, we validated all the values for each individual attribute based on our analysis. We ensure that all text-based feature has correct values while numeric based feature has values that fall within the specific range.

We did analysis using some basic visuals like Histogram, Bar chart to analyse number of unique values exist for each category-based attribute. For numeric based attribute we use violin plot to determine the statistical properties of the attribute. Through this visualization and statistical properties we all ensured if there were no outliers.

4.4 Data Visualization

In a data analysis project, **Data visualization** is yet another important step. Data visualization is a method for visually displaying data using a variety of graphs and charts, which can be useful in identifying trends and making strategic decisions based on data.In this project, we have analyzed data using data visualization. **Matplotlib**, **Seaborn**, and **Plotly** are just a few of the Python libraries we utilize for data visualization. **Microsoft Excel** and business intelligence applications such as **Tableau and PowerBI** were among the other technologies used.

We use a number of visualization techniques, such as the violin plot for outlier detection, the histogram for understanding the distribution of candidate votes, and the bar chart for understanding the proportion of different attributes. Line charts were used to see election trends over time, as well as map visuals for geographical analysis. To generate certain visualizations, we computed new fields, such as the percentage. In a given year, the victorious candidate was elected to that state. To choose the winner, group by function was used. To create a cluster column chart, we also used pivot operations.A variety of techniques were used to shape the dataset to improve visualization. Below shown are a few images that we generated.

1. **Party Performance:**

   This dashboard gives insight into party performance over the entire period. It shows overall party performance throughout the period covered and the details of party won in a particular state throughout the period. It also shows the number of seats won by each party for an year.

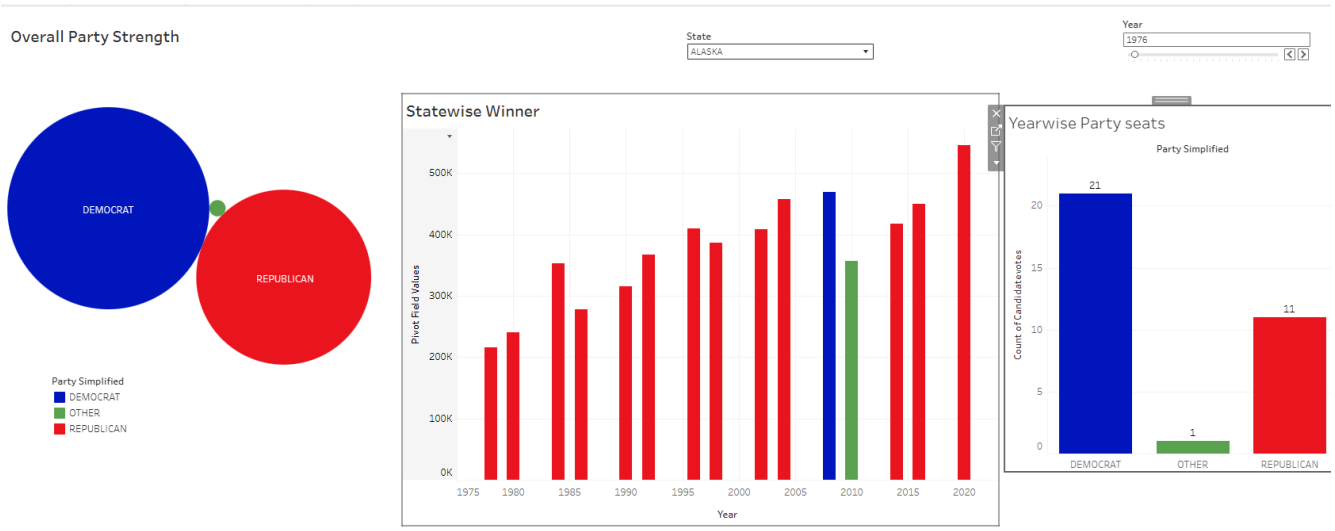

Figure 4: Figure 4: This dashboard illustrates the performance of party over the years.

2. **Party Power Dynamics Over Time:** This dashboard demonstrates the party power dynamics throughout the year. It displays the total votes, and proportion of votes won by each party in separate line charts. It also provides a provision to see the power of a party in a state using state filter option.

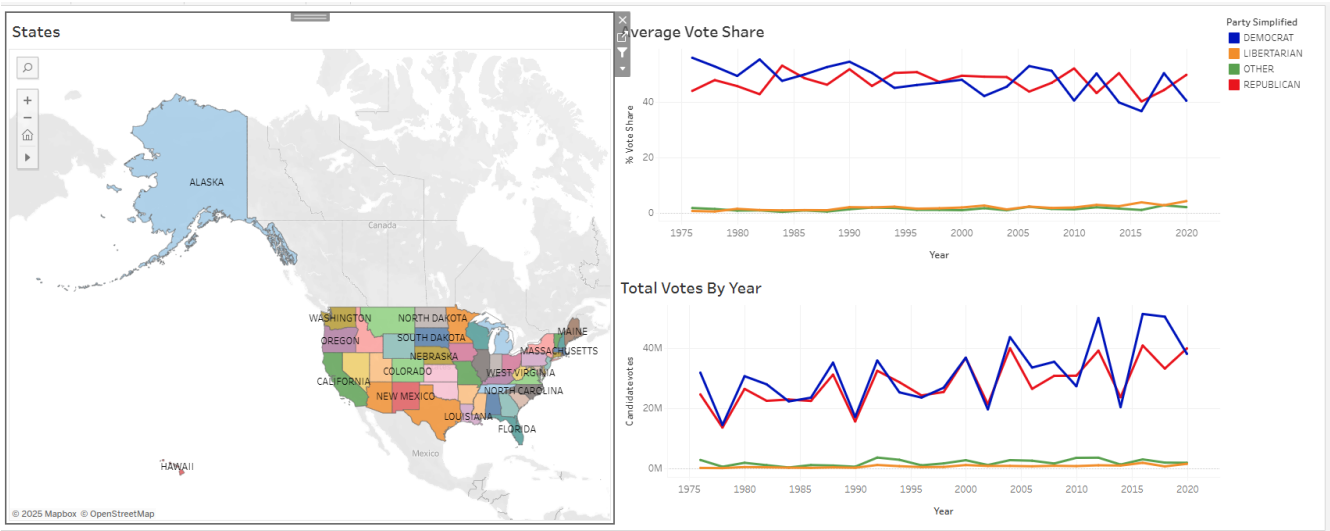

Figure 5: This dashboard demonstrates the election trend for each party over the years.

3. **State Level Election Winners:** This dashboard displays the winner in each state for an year. It displays the winner in map using color encoding. It also shows the percentage of vote for each candidate that gives an insight into winning margin.

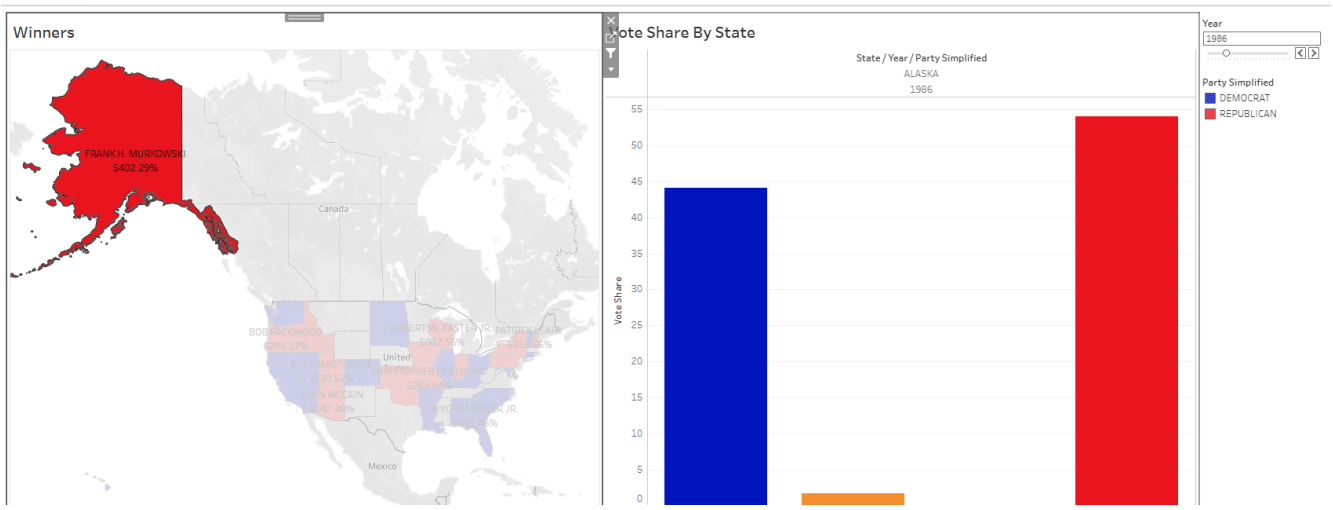

Figure 6: This Dashboard illustrate the Election winner over period of time. This dashboard includes the election winner mapped to Geographical map and Vote share by each party for each state.

4. **Special Election Trends:** This dashboard explains the trend of special election. It displays the details of special election, including total number of states participated in special election conducted in an year in map and graph. It also demonstrates the winner and their winning margin.

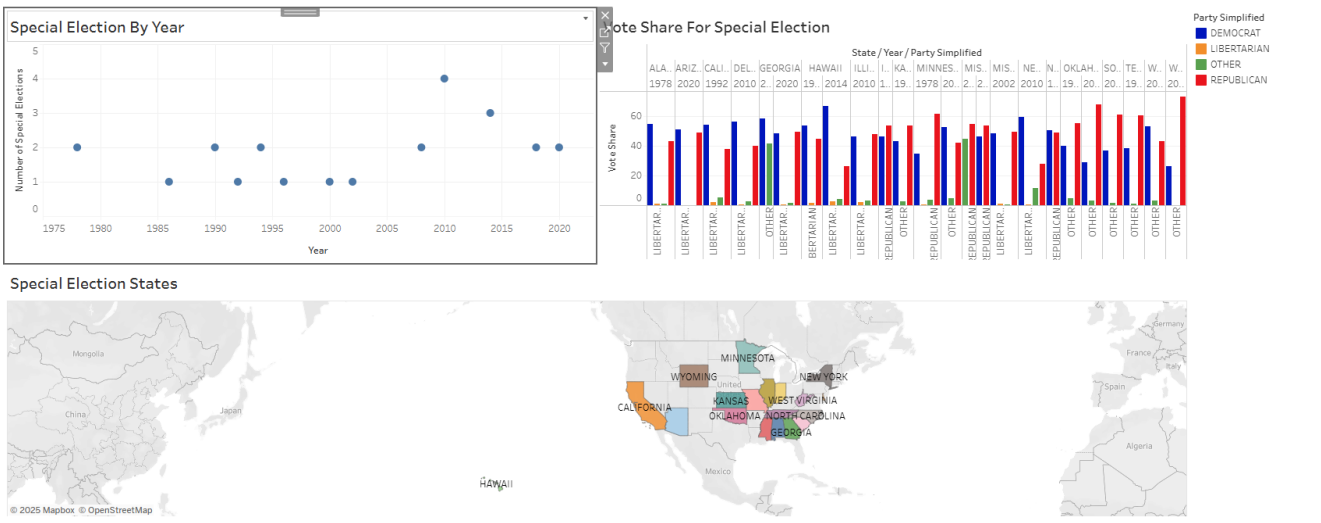

Figure 7: This dashboard illustrates the various visuals related to special elections, which includes Special Election by Year, Vote share for special Election and Special Election stated by map visuals.

4.5 Integration into Web

We have integrated the dashboards created from the preprocessed data into a web-application for public access. Our Web-Application mainly consists of 3 modules.

- **Front-end:** User of the application interacts with the system using this component. This module contains user interfaces that allow public to view and interact with the dashboards. We have used tableu-react npm package[1] that uses Iframe internally to display the dashboards in the system.
- **Application Programming Interface:** This module enables front-end to access the system data. Dashboard details stored in the database tables are fetched using API end points exposed by the Application Programming Interface.
- **Database:** This module stores and manages the whole system data(dashboard details). Our database ensures high performance and reliability and accepts high load simultaneously. It also provides a simple interface for API to fetch the dashboard details.

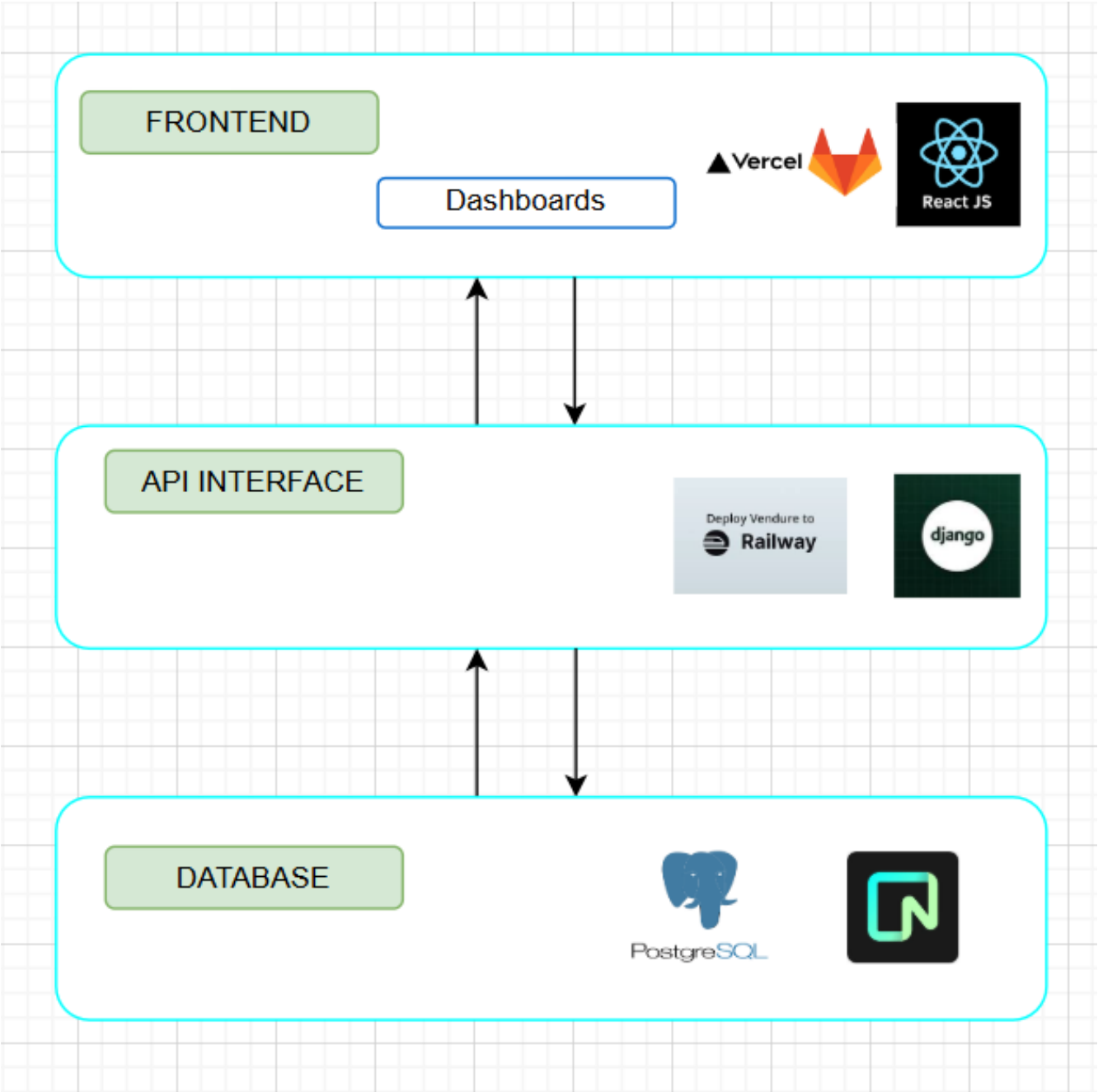

Figure 8: Web-Application Architecture

5 Results and Insights

5.1 Results Identified

- Winners identified for each election corresponds to each year on state wise.
- Election trends identified for each party over the years.
- Special election trends identified.
- We could identify and visualize the power dynamics of party change on year wise as well as state wise.
- We could identify the voting percentage on each year for each party.

5.2 Insights

- There are elections in which more than 2 candidates present for Democratic party in California. In the same election, none of the candidates participated in election from Republican party.
- Our team identified that all the states are not mandated to participate in senate election. There are senate elections in which few states are not participated. We came to know about this scenario while visualization when there were no color visualization for few states. Then we checked the dataset as well as public site to confirm the same.

6 Challenges and Solutions

1. Power BI REST API was our approach to integrate dashboard into React application. However, this requires Power BI pro, premium, or embedded access. We verified it by making API calls from postman.

   Solution:Tableau was our choice for data visualization as we couldn't incorporate Power BI because of compatibility issues. It turned out to be a strong substitute that successfully fulfilled the needs of our project.
2. **Unfamiliarity with Tableau:** We were not familiar with Tableau at beginning.

   Solution: We had to spend time learning how to use it by watching tutorials, practicing with data, and understanding how to create visualizations. It took effort, but it helped us use the tool effectively for our project.
3. **To get proper election data:** Finding accurate and reliable election data was initially challenging, as many sources lacked completeness or consistency.

   Solution: Through extensive research and careful evaluation of various datasets, we were eventually able to identify and obtain a dataset that suited our project needs

7 Discussion and Future Improvement

In this project we have opportunity to learn and explore various Business Intelligence tools, Data Analysis techniques, Data Visualization. Using the Analysis and Visualization we were able to extract many insights.

We do have scope of improvement for the future,

Integrating LLM: We would like to expand this project by integrating Large Language Model which will allow the user to directly ask questions and opportunity to interact with the bot and chatbot will response directly with the answers to the questions.

NPM Package: In this project we have user npm package to integrate react application with Tableau, which provide limited flexibility, therefore, we can create node module package that could smoothly integrate with Tableau in future development.

8 Source Code

1. **Front-end:** https://github.com/athul-narayanan/dashboard$_new$
2. **Back-end:** https://github.com/athul-narayanan/dashboardapi
3. **Data Preprocessing :** Google Collab
4. **Visualizations:**
   - Party Power Dynamics Over Time

- State-Level Election Winners
- Special Election Trends
- Party Performance

9 Web Hosting and Deployment

We have hosted Front-end, Back-end and Database part of the application into free hosting platforms vercel, railway, and neon respectively. Our application can be accessed using the below links.

- **Front-end:** https://dashboard-new-6wpg.vercel.app/
- **Back-end:** https://dashboard-api-production-ceb7.up.railway.app/dashboardapi

10 Acknowledgements

We want to start by thanking our professor Dr. Zamilur Rahman for his advice, helpful criticism, and support, all of which kept us on course and focused. His assistance and knowledge have been essential.

We also like to thank the contributors of the datasets we used, as they were essential to our analysis. Finding and acquiring accurate election data was a difficult procedure, but with persistence and thorough investigation, we were able to obtain a reliable dataset that satisfied our project's requirements.

The constructive feedback, resource sharing, and helpful suggestions throughout the project given by our peers was also helpful. So we are grateful to them aslo. Their advice was crucial to improving our strategy.

This project would not have been achievable,Without the efforts of everyone mentioned, and we sincerely thank them for their support.

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